

developed the language interface pack (LIP), which helps in translating Windows XP into any Indian language. Here you can see the interfaces of the operating system in Indian language. However, the entire operating system is not changed into the regional language but only 20 per cent of the commands that are most used (80 per cent of the time) are changed. To switch back to English, you have to uninstall the LIP, which unfortunately cannot be used like a toggle switch. To promote the Windows OS along with MS Office, Microsoft offers LIP as a free download to registered users.

In the open source world, both OpenOffice and the Linux OS support the Devanagari script.

The latest Windows version, Vista, supports localisation and hence many Indian languages. Vista has introduced some innovative features like the multilingual user interface (.mui) files. These files contain the local language resources. MUI works in two parts, one of which is the OS and the other is the language. Both work independent of each other. Thus, when you update Vista, the language part is not disturbed. This is a major change in the localisation of Microsoft's OS. In earlier versions of Windows, the local language was superimposed on a basically English OS. With this improvement, it has become easier to apply Indian languages to Microsoft products.

In the open source world, both OpenOffice and the Linux OS support the Devanagari script. On the www.indlinux.org website, you can download the Rangoli 1.0 beta version (690 MB), which has Hindi, Bengali, Tamil, Gujarati and Punjabi interfaces based on Gnome 2.10 and KDE 3.4.3.

C-DAC has a project called IndiX that deals with bringing Indian lan-

guages to the Linux OS. IndiX is the compilation of Indian languages for GNU/Linux. It supports Indian languages like Assamese, Bengali, Devanagari, Gujarati, Gurmukhi, Maithili, Malayalam, Tamil, Oriya, Telugu and Kannada. You can run the IndiX project CD directly without installing it on your hard disk. This CD supports KDE 3.2 as the standard desktop, the Mozilla Suite and X Multimedia System.

Office products in Indian languages

Microsoft's website www.bhashaindia.com has been developed to support Indian languages. From this site you can download LIPs for different Indian languages. After Microsoft adopted Unicode fonts, things have become easier. Now, after doing some minor changes in the set-up of your personal computer, you can use it to type Indian languages.

In addition to this support to Indic languages, Microsoft has brought out an exclusive Hindi Office suite, which uses a Hindi interface, and one can easily navigate the menus and toolbars, which are in Hindi. It offers multiple keyboard options like Inscript, Transliteration, Remington and Godrej. Microsoft is trying to enhance the capabilities of this office package by trying to include a Hindi text-to-speech converter, extra fonts and some dictionaries.

The latest available version is Office 2003. Microsoft has dedicated a separate website for Hindi Office at www.microsoft.com/india/officehindi/, where you can get all the information about it.

Leap Office from C-DAC is the complete Indian language software for office applications. This package includes a multilingual spellchecker, 'find' and 'replace' functions, an online official language dictionary, etc. It has an online active keyboard for those who do not know how to type in Indian languages.

iLEAP from C-DAC is an intelli-

gent, Internet-ready, Indian-language wordprocessor, in which you can switch between languages very easily. You can send e-mails written in Indian languages and create webpages in Indian languages. It also has a multilingual editor.

E-mail and search engine sites

There are many e-mail websites that support most of the Indian languages. Examples are www.email2india.com/ (supports 13 Indian languages), <http://mail.webdunia.com/> (supports nine Indian languages) and www.gmail.com. You can even chat in an Indian language on many websites using the transliteration technique.

Apart from Google, which provides facilities to search in almost nine Indian languages, there are search engines available for Indian languages. www.raftaar.in is useful to search the world wide web in Hindi, using the Devanagari script. Results are displayed in Hindi only. www.guruji.com is a search engine for seven Indian languages—Hindi, Telugu, Marathi, Kannada, Malayalam, Gujarati and Tamil. Users operate a virtual keyboard to type in their respective languages, and then the results of their search are displayed in that language.

Online tutorials

LILA is a software developed by C-DAC for this purpose. It stands for 'learning Indian languages through artificial intelligence.' It can even help one learn Hindi using one of the South Indian languages.

LILA, an intelligent self-tutoring system, is specially designed for government employees working in banks and other public-sector undertakings. There is even a Web version of LILA, in which online tutorials for five Indian languages (Hindi, Gujarati, Telugu, Tamil and Bengali) are available. ●

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